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# Initial Solution V2

## Comparative Simulation Report

Task replacement recourse, paired experiment এবং ফলাফল

|                    |                                          |
|--------------------|------------------------------------------|
| Project:           | PGBM thesis and simulation work          |
| সমাধানের ধরন:      | Formulation aligned one-for-one recourse |
| Environment:       | Synthetic এবং DU outdoor                 |
| মোট configuration: | ২৪০টি paired configuration, ৪৮০টি row    |
| Seed:              | ১০১-১১০                                  |
| Report date:       | ২৮ সেপ্টেম্বর ২০২৬                       |

এই report-এ V2-এর core recourse rule, runtime state, paired experiment এবং V1 বনাম V2 ফলাফল একই flow-তে ব্যাখ্যা করা হয়েছে।

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## 1 V2-এর core feature

V2 হলো formulation-aligned one-for-one active-mission task replacement recourse। নতুন task আসলে সব responder UAV active থাকলে প্রতিটি active mission-এর uncompleted task একবার করে replace করার candidate পরীক্ষা করা হয়। Completed task এবং completed route prefix অপরিবর্তিত থাকে। New task-এর item demand current onboard inventory দিয়ে মেনানো সম্ভব হতে হয়; remaining route existing collision-aware 3D routing layer দিয়ে একই base-এ ফেরত তৈরি হয়; এবং শুধু positive  $\Delta J = J_{\text{new}} - J_{\text{old}}$  হলে সর্বোচ্চ gain-এর candidate গ্রহণ করা হয়। Displaced task waiting queue-তে ফিরে যায়।

- One-for-one replacement: একটি active UAV এবং একটি uncompleted task।
- Completed prefix protection: completed delivery এবং route segment পরিবর্তন করা হয় না।
- Current onboard inventory: reserve inventory বা base reload V2-তে নেই।
- Remaining route recourse: current position থেকে task suffix এবং একই base-এ return route নতুন করে তৈরি হয়।
- Positive service-value gain:  $\Delta J > 0$  না হলে active mission অপরিবর্তিত থাকে।
- V1 isolation: V1 entry point এবং canonical evidence overwrite করা হয়নি।

Reserve inventory, base reload, multiple-task replacement এবং full fleet reoptimization Version 3 scope।

## 2 Experiment-এর সংক্ষিপ্ত চিত্র

V2 একই ১২০ মিনিটের seeded event-driven protocol ব্যবহার করে: ০-৪০ মিনিট high, ৪০-৮০ মিনিট medium এবং ৮০-১২০ মিনিট low arrival phase। দুইটি environment, ৩০/৬০/৯০ task load, ৩/৫/৮/১০ UAV এবং ১০টি seed (১০১-১১০) ব্যবহার করা হয়েছে। ফলে ২৪০টি configuration-এ V1 এবং V2-এর ৪৮০টি metrics row এবং ২৪০টি V2 decision trace তৈরি হয়েছে। V1-এর preserved canonical matrix comparison row হিসেবে রাখা হয়েছে। V2 একই local scene এবং task seed protocol-এ চালানো হয়েছে। In-memory paired validation-এ recourse disabled করলে V2 fixed control-এর objective, served task, distance, energy এবং candidate count V1-এর সঙ্গে মিলে গেছে।

## 3 V2 algorithm কীভাবে assignment এবং recourse ঠিক করে

V2-এর initial dispatch V1-এর bounded brute-force assignment এবং task-order rule ব্যবহার করে। নতুন task arrival হলে V2 পুরো fleet replan করে না। Current mission state থেকে প্রতিটি active UAV এবং প্রতিটি uncompleted task-এর জন্য একটি candidate তৈরি হয়।

1. Current time-এ UAV-এর position, completed task, remaining task, onboard item, payload এবং consumed energy project করা হয়।
2. একটি uncompleted task সরিয়ে নতুন task-টি একই mission position-এ বসানো হয়।
3. Service চলতে থাকলে তা interrupt করা হয় না; candidate current service শেষ হওয়ার পরের suffix পরিবর্তন করে।
4. New task এবং revised mission-এর item demand current onboard inventory দিয়ে মেনানো যায় কি না পরীক্ষা করা হয়।
5. Current position থেকে revised task sequence এবং একই physical base পর্যন্ত collision-aware 3D route তৈরি হয়।
6. Payload, route, energy reserve এবং ১২০ মিনিটের horizon feasibility পরীক্ষা হয়।
7. Old remaining value, new remaining value এবং  $\Delta J$  হিসাব করা হয়।
8. সর্বোচ্চ positive  $\Delta J$  গ্রহণ করা হয়; positive candidate না থাকলে নতুন task queue-তে থাকে।

Final episode objective প্রত্যেক completed task-এর service value একবার গণনা করে; replanning snapshot যোগ করে double counting করা হয় না।

## 4 Simulation environment এবং route flow

V2 একই V1 scene, task, parcel, drop-off waypoint, payload, energy এবং route contracts ব্যবহার করে। প্রতিটি task individual delivery request; task-এর food, water এবং medical demand, ২-৪ m vertical drop-off waypoint এবং ০.৫ মিনিট per parcel service duration অপরিবর্তিত আছে। একটি active mission-এর route হলো Base → task waypoint sequence → Base। Recourse trigger হলে completed route অংশকে আবার plan করা হয় না। UAV-এর projected current position থেকে শুধু remaining suffix নতুন করে route করা হয়। প্রতিটি leg existing cruise-altitude check, direct line check, blocked হলে grid A\* এবং vertical clearance check অনুসরণ করে।

## 5 Result বোঝার জন্য প্রধান metric

Objective completion-time exponential service value; served task এবং dropped parcel আলাদা unit; accepted replacement positive  $\Delta J$ -এর count; replacement gain event-level evidence; runtime, planner, route এবং recourse সময়ের আলাদা breakdown।

## 6 Environment-level V1 বনাম V2 ফলাফল

সারণী 1: Environment-level V1 এবং V2 aggregate result।

| Environment | Algorithm | Objective    | Served        | Task rate | Parcel rate | Deferred      | Delay min    | Runtime s |
|-------------|-----------|--------------|---------------|-----------|-------------|---------------|--------------|-----------|
| DU outdoor  | V1        | 10.43 ± 4.22 | 20.12 ± 7.95  | 33.5%     | 36.6%       | 39.88 ± 24.05 | 31.64 ± 4.05 | 20.225    |
| DU outdoor  | V2        | 11.70 ± 4.53 | 19.63 ± 7.63  | 32.7%     | 33.4%       | 40.37 ± 23.82 | 23.21 ± 3.97 | 59.408    |
| Synthetic   | V1        | 23.48 ± 9.36 | 40.40 ± 16.73 | 67.3%     | 69.1%       | 19.60 ± 19.34 | 18.30 ± 4.54 | 3.733     |
| Synthetic   | V2        | 23.84 ± 9.34 | 39.36 ± 16.19 | 65.6%     | 66.9%       | 20.64 ± 19.75 | 16.08 ± 3.33 | 10.355    |

এই table থেকে কয়েকটি ফলাফল পরীক্ষার। DU outdoor-এ V2 objective 10.43 থেকে 11.70 হয়েছে (প্রায় 12.2% বৃদ্ধি), যদিও served task 20.12 থেকে 19.63 এবং parcel rate 36.6% থেকে 33.4%-এ সামান্য কমেছে। Synthetic-এ objective 23.48 থেকে 23.84 (প্রায় 1.5%) বেড়েছে, কিন্তু served task এবং parcel rate-ও সামান্য কমেছে। অর্থাৎ V2-এর লাভ raw task count নয়; active mission-এর মধ্যে কোন task আগে complete করলে time-sensitive service value বেশি হবে, সেটি বেছে নেওয়া। সবচেয়ে গুরুত্বপূর্ণ পরিবর্তনটি Delay min-এ। DU outdoor-এ average delay 31.64 থেকে 23.21 মিনিটে (8.43 মিনিট বা প্রায় 26.6%) এবং Synthetic-এ 18.30 থেকে 16.08 মিনিটে (2.22 মিনিট বা প্রায় 12.1%) কমেছে। Delay min হলো task detect হওয়ার সময় থেকে delivery complete হওয়া পর্যন্ত গড় সময়। নতুন task এলে V2 শুধু সেই replacement নেয় যার revised remaining mission value পুরনো mission-এর চেয়ে বেশি; ফলে অপেক্ষমাণ high-value task অনেক ক্ষেত্রে আগের completion position পায়। তবে এটি সব task-এর delay কমেছে-এমন দাবি নয়; কিছু lower-value displaced task queue-তে ফেরত যায়, তাই served count কমেও average completed-task delay কমেতে পারে। এর বিপরীতে V2 runtime DU outdoor-এ 20.225 থেকে 59.408 s এবং Synthetic-এ 3.733 থেকে 10.355 s হয়েছে। কারণ প্রতিটি recourse trigger-এ সম্ভাব্য active UAV ও displaced task-এর জন্য feasibility এবং revised route পরীক্ষা করা হয়। তাই V2-এর মূল trade-off হলো-priority-weighted service value এবং completion delay উন্নত করার বিনিময়ে অতিরিক্ত computation।

## 7 Task load এবং UAV sensitivity

Task load বাড়লে queue pressure বাড়ে এবং V1 ও V2 দুই version-এর service rate সাধারণত কমে। UAV সংখ্যা বাড়লে available capacity বাড়ে, কিন্তু নতুন task idle UAV-তে dispatch হয়ে গেলে recourse trigger-এর সুযোগ কমেতেও পারে। তাই V2 accepted replacement fleet size-এর সঙ্গে monotonic হওয়া আবশ্যিক নয়।

সারণী ২: সম্পূর্ণ V1 বনাম V2 task load এবং fleet matrix।

| Environment | Task | UAV | V1 Obj | V2 Obj | $\Delta$ Obj | V1 Served | V2 Served | $\Delta$ Served | V1 Parcel | V2 Parcel | $\Delta$ Parcel | V2 Repl | V2 Runtime |
|-------------|------|-----|--------|--------|--------------|-----------|-----------|-----------------|-----------|-----------|-----------------|---------|------------|
| DU outdoor  | 30   | 3   | 5.11   | 5.92   | +0.81        | 10.50     | 10.20     | -0.30           | 30.9%     | 26.9%     | -4.0 pp         | 6.10    | 37.612     |
| DU outdoor  | 30   | 5   | 7.47   | 8.32   | +0.85        | 15.90     | 15.40     | -0.50           | 48.1%     | 43.7%     | -4.4 pp         | 7.00    | 42.588     |
| DU outdoor  | 30   | 8   | 9.87   | 10.23  | +0.35        | 20.20     | 19.60     | -0.60           | 63.8%     | 61.4%     | -2.4 pp         | 4.70    | 38.111     |
| DU outdoor  | 30   | 10  | 10.59  | 10.80  | +0.22        | 21.20     | 20.80     | -0.40           | 67.2%     | 65.2%     | -2.0 pp         | 3.20    | 28.559     |
| DU outdoor  | 60   | 3   | 5.95   | 7.42   | +1.48        | 10.80     | 11.40     | +0.60           | 16.5%     | 14.4%     | -2.2 pp         | 8.00    | 50.711     |
| DU outdoor  | 60   | 5   | 9.49   | 10.70  | +1.21        | 17.80     | 17.20     | -0.60           | 27.6%     | 23.4%     | -4.3 pp         | 9.10    | 54.161     |
| DU outdoor  | 60   | 8   | 13.63  | 14.77  | +1.14        | 26.90     | 25.30     | -1.60           | 42.4%     | 37.8%     | -4.6 pp         | 13.00   | 50.260     |
| DU outdoor  | 60   | 10  | 16.20  | 17.10  | +0.90        | 32.60     | 30.70     | -1.90           | 52.5%     | 47.4%     | -5.1 pp         | 13.00   | 36.383     |
| DU outdoor  | 90   | 3   | 6.26   | 7.55   | +1.30        | 11.20     | 10.90     | -0.30           | 11.2%     | 9.4%      | -1.9 pp         | 9.70    | 65.505     |
| DU outdoor  | 90   | 5   | 9.80   | 11.80  | +1.99        | 17.60     | 17.50     | -0.10           | 18.3%     | 15.9%     | -2.5 pp         | 14.00   | 61.833     |
| DU outdoor  | 90   | 8   | 14.08  | 16.47  | +2.39        | 25.60     | 25.50     | -0.10           | 27.3%     | 24.6%     | -2.7 pp         | 17.40   | 113.128    |
| DU outdoor  | 90   | 10  | 16.76  | 19.35  | +2.59        | 31.20     | 31.10     | -0.10           | 33.5%     | 31.2%     | -2.2 pp         | 17.10   | 134.039    |
| Synthetic   | 30   | 3   | 11.98  | 12.63  | +0.65        | 22.60     | 22.00     | -0.60           | 71.1%     | 67.5%     | -3.7 pp         | 6.40    | 4.040      |
| Synthetic   | 30   | 5   | 15.30  | 15.32  | +0.01        | 27.20     | 27.00     | -0.20           | 90.1%     | 89.0%     | -1.2 pp         | 3.70    | 3.222      |
| Synthetic   | 30   | 8   | 16.26  | 16.28  | +0.02        | 27.50     | 27.50     | +0.00           | 90.9%     | 90.9%     | +0.0 pp         | 0.80    | 3.355      |
| Synthetic   | 30   | 10  | 16.44  | 16.44  | +0.00        | 27.50     | 27.50     | +0.00           | 90.9%     | 90.9%     | +0.0 pp         | 0.00    | 3.612      |
| Synthetic   | 60   | 3   | 15.65  | 16.40  | +0.76        | 25.40     | 24.70     | -0.70           | 37.3%     | 34.6%     | -2.6 pp         | 8.80    | 11.164     |
| Synthetic   | 60   | 5   | 22.84  | 23.71  | +0.87        | 39.90     | 38.60     | -1.30           | 60.9%     | 57.8%     | -3.1 pp         | 11.00   | 14.512     |
| Synthetic   | 60   | 8   | 29.70  | 29.78  | +0.08        | 53.30     | 51.90     | -1.40           | 87.6%     | 84.6%     | -3.0 pp         | 9.70    | 17.878     |
| Synthetic   | 60   | 10  | 31.86  | 31.77  | -0.09        | 55.30     | 54.50     | -0.80           | 91.6%     | 89.6%     | -2.0 pp         | 5.40    | 11.775     |
| Synthetic   | 90   | 3   | 17.81  | 18.34  | +0.54        | 27.70     | 26.30     | -1.40           | 25.2%     | 22.8%     | -2.3 pp         | 13.30   | 8.992      |
| Synthetic   | 90   | 5   | 26.11  | 26.73  | +0.62        | 41.80     | 40.20     | -1.60           | 40.3%     | 38.0%     | -2.3 pp         | 15.50   | 11.353     |
| Synthetic   | 90   | 8   | 36.28  | 37.00  | +0.72        | 62.30     | 60.50     | -1.80           | 64.2%     | 61.2%     | -3.1 pp         | 14.40   | 16.359     |
| Synthetic   | 90   | 10  | 41.52  | 41.65  | +0.12        | 74.30     | 71.60     | -2.70           | 79.2%     | 75.6%     | -3.6 pp         | 13.80   | 17.993     |

এই matrix-এর প্রধান patternগুলো হলো: (১) DU outdoor-এর ১২টি configuration-এর সবকটিতেই V2 objective V1-এর চেয়ে বেশি। বিশেষ করে ৯০ task-এ gain 1.30 থেকে 2.59 পর্যন্ত, অর্থাৎ load বাড়লে recourse-এর priority benefit বেশি দৃশ্যমান হয়। (২) DU outdoor-এ ৬০ task/৩ UAV ছাড়া served task সামান্য কমেছে বা প্রায় অপরিবর্তিত থেকেছে; parcel rate-ও সব cell-এ কমেছে। কারণ V2 কিছু lower-value active task সরিয়ে বেশি time-sensitive task বসায়-এটি throughput-maximization rule নয়। (৩) Synthetic-এ low load এবং বেশি UAV থাকলে V1-এর capacity প্রায় যথেষ্ট; ৩০ task/১০ UAV-এ accepted replacement গড় ০ এবং objective পরিবর্তন ০.০০। তাই সেখানে V2-এর বাড়তি লাভ সীমিত। (৪) Synthetic-এর ৬০ task/১০ UAV cell-এ একমাত্র সামান্য negative objective change (-0.09) দেখা গেছে-এই configuration-এ replacement-এর লাভ computation/queue trade-off পুরোপুরি offset করতে পারেনি। (৫) ৯০ task-এ accepted replacement এবং runtime সাধারণত বাড়ে; DU outdoor-এ ৯০ task/১০ UAV runtime 134.039 s পর্যন্ত উঠেছে। সুতরাং matrix দেখায় যে V2-এর value সবচেয়ে বেশি congestion বা active-mission overlap থাকা configuration-এ। UAV বাড়লে service capacity বাড়ে, কিন্তু নতুন task idle UAV-তে সরাসরি dispatch হলে replacement দরকার কমে; তাই accepted replacement fleet size-এর সঙ্গে সবসময় monotonic হয় না।

## 8 Recourse decision এবং rejection evidence

প্রতিটি trigger-এর candidate UAV, displaced task, item feasibility, route feasibility, energy feasibility, old value, new value, gain এবং final decision decision trace-এ রাখা হয়েছে। নিচের table-এ trigger-level rejection reason দেওয়া হলো; candidate-level কারণ সম্পূর্ণ JSON trace-এ দেখা যাবে। এই অংশটির সহজ অর্থ হলো: নতুন

| Trigger-level rejection reason | Evidence count |
|--------------------------------|----------------|
| no_positive_gain               | 2113           |
| no_feasible_candidate          | 1842           |
| no_uncompleted_active_task     | 433            |

task আসার সময় যদি UAV-গুলো active থাকে, V2 দেখে কোনো চলমান mission-এর একটি uncompleted task সরিয়ে নতুন task বসালে সত্যিই লাভ হবে কি না। একটি trigger-এ একাধিক candidate পরীক্ষা হতে পারে; তাই table-এর সংখ্যা candidate count নয়, trigger-level final outcome। মোট ৬৬৩৯টি trigger-এর মধ্যে ২২৫১টি replacement গৃহীত এবং ৪৩৮৮টি গৃহীত হয়নি-অর্থাৎ acceptance rate প্রায় 33.9%। no\_positive\_gain (2113) সবচেয়ে বেশি দেখা গেছে। অর্থাৎ candidate route ও item-এর দিক থেকে সম্ভব হলেও পুরনো task সরিয়ে নতুন task বসালে remaining service value বাড়েনি; তাই বর্তমান mission অপরিবর্তিত রাখা হয়েছে। no\_feasible\_candidate (1842) মানে কোনো candidate-ই item/inventory, route, energy reserve, return বা horizon-এর সব শর্ত একসঙ্গে পূরণ করতে পারেনি। no\_uncompleted\_active\_task (433) মানে active mission-এ সরানোর মতো uncompleted task ছিল না। এই rejection-এ নতুন task হারিয়ে যায় না-pending queue-তে থেকে পরবর্তী dispatch-এর জন্য অপেক্ষা করে। Candidate-level কারণগুলো JSON decision trace-এ আলাদা করে রাখা হয়েছে।

## 9 Runtime এবং প্রধান bottleneck

এখানে Runtime s বলতে simulation program-এর computation time বোঝায়; এটি UAV-এর real flight time বা task delay নয়। Planner s হলো initial assignment search-এর সময়, Route s হলো collision-aware route তৈরি/মূল্যায়নের সময়, আর Recourse s হলো V2-তে নতুন task replacement candidate পরীক্ষা করার অতিরিক্ত সময়।

| Environment | Algorithm | Runtime s | Planner s | Route s | Recourse s | Candidate | Truncated | Safe return |
|-------------|-----------|-----------|-----------|---------|------------|-----------|-----------|-------------|
| DU outdoor  | V1        | 20.225    | 20.223    | 19.566  | 0.000      | 249477    | 4.1       | 100.0%      |
| DU outdoor  | V2        | 59.408    | 46.568    | 44.692  | 12.784     | 288639    | 4.8       | 100.0%      |
| Synthetic   | V1        | 3.733     | 3.731     | 3.561   | 0.000      | 65211     | 1.0       | 100.0%      |
| Synthetic   | V2        | 10.355    | 8.961     | 8.513   | 1.326      | 57343     | 0.8       | 100.0%      |

Table-এর প্রধান bottleneck হলো route evaluation। V1-এ DU outdoor-এর 20.225 s runtime-এর মধ্যে route অংশ 19.566 s (প্রায় 96.7%); Synthetic-এ 3.561 s (প্রায় 95.4%)। তবে V2-এর Runtime, Planner, Route এবং Recourse timer-গুলো mutually exclusive নয়। Planner-এর মধ্যে route evaluation থাকতে পারে এবং recourse candidate যাচাইয়ের সময়ও route check চলে; তাই এগুলো সরাসরি যোগ করে total runtime ধরা যাবে না। উদাহরণ হিসেবে DU outdoor V2-তে Route 44.692 s + Recourse 12.784 s = 57.476 s, যেখানে total Runtime 59.408 s; বাকি 1.932 s event handling, dispatch bookkeeping এবং অন্যান্য overhead। Synthetic-এ 8.513 + 1.326 = 9.839 s, total 10.355 s; residual 0.516 s। অর্থাৎ breakdown-টি bottleneck বোঝার diagnostic view, additive accounting নয়। এই ফল দেখায় V2-এর বাড়তি computation-এর বড় অংশ repeated route feasibility পরীক্ষা, আর underlying route search-ই এখনও প্রধান cost centre। DU outdoor Synthetic-এর তুলনায় অনেক ধীর,

কারণ DU-তে polygon obstacle geometry এবং blocked হলে grid A\* path search বেশি কাজ করে। V2 runtime V1-এর তুলনায় DU-তে প্রায় 2.94 গুণ এবং Synthetic-এ প্রায় 2.77 গুণ হয়েছে। Safe return 100% মানে simulation-এর feasibility filter পেরিয়ে সব executed mission base-এ ফিরেছে; এটি field-flight reliability-এর প্রমাণ নয়। ‘Search truncated’ হলো যেসব dispatch ৫০,০০০ candidate limit-এ পৌঁছেছে তার গড় সংখ্যা, আর Candidate হলো পরীক্ষিত candidate plan-এর গড় count। তাই পরবর্তী efficiency কাজের প্রধান দিক হবে route reuse/cache hit measurement, early feasibility pruning এবং scalable candidate search।

## 10 Energy model এবং energy ফলাফল

V1 এবং V2 উভয় version-এ energy একটি configured accounting model দিয়ে গণনা করা হয়েছে। কোনো travel segment-এর জন্য  $d$  হলো route distance,  $m$  হলো বহন করা payload mass এবং  $a^+$  হলো positive ascent distance।

$$E_{travel} = d e_{distance} + m e_{payload} + a^+ e_{ascent}, \quad E_{service} = \tau e_{service}, \quad E_{mission} = \sum E_{travel} + \sum E_{service}.$$

এই run-এ  $e_{distance} = 1.0$  J/m,  $e_{payload} = 0.05$  J/(kg m),  $e_{ascent} = 0.20$  J/m এবং  $e_{service} = 0.50$  J/min। প্রতিটি initial বা recourse candidate-এর জন্য UAV-এর ইতিমধ্যে ব্যবহৃত energy-এর সঙ্গে revised route-এর additional energy যোগ করে check করা হয়:  $E_{consumed} + E_{additional} + E_{reserve} \leq E_{capacity}$ । Capacity 1000 J এবং reserve 200 J; তাই projected remaining energy-এর অন্তত reserve অংশ mission শেষে অবশিষ্ট থাকতে হয়। Total energy পুরো episode-এ সম্পন্ন বা horizon পর্যন্ত চলা সব mission-এর cumulative energy। এটি কোনো একক UAV-এর একবারের battery state নয়। প্রতিটি UAV base-এ ফিরে resupply/turnaround শেষ করলে পরের mission-এ নতুন full capacity থেকে শুরু করে; V1/V2-তে battery degradation বা mission-to-mission persistent battery depletion model করা হয়নি।

| Environment | Algorithm | Energy J        | Energy/task J | Energy/parcel J | Distance m      | Travel min    |
|-------------|-----------|-----------------|---------------|-----------------|-----------------|---------------|
| DU outdoor  | V1        | 6142.4 ± 2417.8 | 310.2 ± 62.2  | 199.3 ± 38.8    | 5928.2 ± 2335.5 | 592.8 ± 233.5 |
| DU outdoor  | V2        | 6078.6 ± 2385.3 | 314.7 ± 69.0  | 224.5 ± 61.3    | 5854.1 ± 2300.6 | 585.4 ± 230.1 |
| Synthetic   | V1        | 5319.7 ± 2101.8 | 134.4 ± 28.4  | 84.3 ± 17.9     | 5041.4 ± 1988.8 | 504.1 ± 198.9 |
| Synthetic   | V2        | 5368.8 ± 2123.8 | 138.9 ± 27.6  | 88.5 ± 17.0     | 5083.6 ± 2006.9 | 508.4 ± 200.7 |

Energy result-এর মূল trade-off হলো: DU outdoor-এ V2 total energy 6142.4 J থেকে 6078.6 J-এ এবং distance 5928.2 m থেকে 5854.1 m-এ কমেছে, কারণ V2 কিছু lower-value task replace বা queue করে এবং infeasible বা energy-expensive candidate গ্রহণ করে না। কিন্তু completed task ও parcel কম হওয়ায় প্রতি served task energy 310.2 থেকে 314.7 J এবং প্রতি dropped parcel 199.3 থেকে 224.5 J হয়েছে। Synthetic-এ V2 total energy 5319.7 J থেকে 5368.8 J এবং distance 5041.4 m থেকে 5083.6 m-এ সামান্য বেড়েছে; accepted recourse-এর revised route এই অতিরিক্ত movement তৈরি করেছে। তাই শুধু total energy দিয়ে efficiency বিচার করা যাবে না; cumulative energy, energy per completed service এবং distance একসঙ্গে দেখতে হবে।

## 11 Expanded V1 বনাম V2 result matrix

নিচের দুইটি matrix-এ প্রতিটি cell-এর ১০টি seed-এর mean ± standard deviation দেওয়া হয়েছে। ফলে objective, served task, parcel volume, queue, energy, recourse এবং runtime-এর seed variability একই সঙ্গে দেখা যায়।

সারণী 3: Service, parcel এবং queue metrics: mean  $\pm$  standard deviation।

| Environment | Task | UAV | Alg | Obj              | Served           | Task rate         | Req. parcel     | Drop parcel    | Deferred         | Delay            | Peak queue     |
|-------------|------|-----|-----|------------------|------------------|-------------------|-----------------|----------------|------------------|------------------|----------------|
| DU outdoor  | 30   | 3   | V1  | 5.11 $\pm$ 1.50  | 10.50 $\pm$ 3.10 | 35.0% $\pm$ 10.3% | 48.8 $\pm$ 3.9  | 14.9 $\pm$ 2.2 | 19.50 $\pm$ 3.10 | 34.33 $\pm$ 3.96 | 19.5 $\pm$ 3.1 |
| DU outdoor  | 30   | 3   | V2  | 5.92 $\pm$ 1.24  | 10.20 $\pm$ 2.82 | 34.0% $\pm$ 9.4%  | 48.8 $\pm$ 3.9  | 13.0 $\pm$ 3.8 | 19.80 $\pm$ 2.82 | 24.28 $\pm$ 7.21 | 19.8 $\pm$ 2.8 |
| DU outdoor  | 30   | 5   | V1  | 7.47 $\pm$ 1.42  | 15.90 $\pm$ 3.31 | 53.0% $\pm$ 11.0% | 48.8 $\pm$ 3.9  | 23.3 $\pm$ 4.4 | 14.10 $\pm$ 3.31 | 33.31 $\pm$ 5.01 | 14.3 $\pm$ 3.3 |
| DU outdoor  | 30   | 5   | V2  | 8.32 $\pm$ 1.35  | 15.40 $\pm$ 3.13 | 51.3% $\pm$ 10.4% | 48.8 $\pm$ 3.9  | 21.1 $\pm$ 4.7 | 14.60 $\pm$ 3.13 | 26.05 $\pm$ 4.38 | 14.8 $\pm$ 3.2 |
| DU outdoor  | 30   | 8   | V1  | 9.87 $\pm$ 1.57  | 20.20 $\pm$ 3.12 | 67.3% $\pm$ 10.4% | 48.8 $\pm$ 3.9  | 31.0 $\pm$ 6.1 | 9.80 $\pm$ 3.12  | 28.36 $\pm$ 3.05 | 10.2 $\pm$ 3.0 |
| DU outdoor  | 30   | 8   | V2  | 10.23 $\pm$ 1.68 | 19.60 $\pm$ 3.63 | 65.3% $\pm$ 12.1% | 48.8 $\pm$ 3.9  | 29.8 $\pm$ 7.4 | 10.40 $\pm$ 3.63 | 25.66 $\pm$ 2.68 | 10.7 $\pm$ 3.5 |
| DU outdoor  | 30   | 10  | V1  | 10.59 $\pm$ 1.46 | 21.20 $\pm$ 2.74 | 70.7% $\pm$ 9.1%  | 48.8 $\pm$ 3.9  | 32.8 $\pm$ 6.5 | 8.80 $\pm$ 2.74  | 25.97 $\pm$ 3.18 | 9.2 $\pm$ 2.4  |
| DU outdoor  | 30   | 10  | V2  | 10.80 $\pm$ 1.45 | 20.80 $\pm$ 2.86 | 69.3% $\pm$ 9.5%  | 48.8 $\pm$ 3.9  | 31.8 $\pm$ 7.1 | 9.20 $\pm$ 2.86  | 24.88 $\pm$ 2.78 | 9.5 $\pm$ 2.6  |
| DU outdoor  | 60   | 3   | V1  | 5.95 $\pm$ 1.27  | 10.80 $\pm$ 1.62 | 18.0% $\pm$ 2.7%  | 101.3 $\pm$ 5.1 | 16.7 $\pm$ 1.5 | 49.20 $\pm$ 1.62 | 33.64 $\pm$ 3.56 | 49.2 $\pm$ 1.6 |
| DU outdoor  | 60   | 3   | V2  | 7.42 $\pm$ 1.11  | 11.40 $\pm$ 1.71 | 19.0% $\pm$ 2.9%  | 101.3 $\pm$ 5.1 | 14.5 $\pm$ 2.1 | 48.60 $\pm$ 1.71 | 21.94 $\pm$ 3.70 | 48.6 $\pm$ 1.7 |
| DU outdoor  | 60   | 5   | V1  | 9.49 $\pm$ 1.52  | 17.80 $\pm$ 1.99 | 29.7% $\pm$ 3.3%  | 101.3 $\pm$ 5.1 | 27.9 $\pm$ 2.1 | 42.20 $\pm$ 1.99 | 33.21 $\pm$ 4.06 | 42.2 $\pm$ 2.0 |
| DU outdoor  | 60   | 5   | V2  | 10.70 $\pm$ 1.55 | 17.20 $\pm$ 2.49 | 28.7% $\pm$ 4.1%  | 101.3 $\pm$ 5.1 | 23.6 $\pm$ 3.2 | 42.80 $\pm$ 2.49 | 23.16 $\pm$ 1.62 | 42.8 $\pm$ 2.5 |
| DU outdoor  | 60   | 8   | V1  | 13.63 $\pm$ 2.00 | 26.90 $\pm$ 3.03 | 44.8% $\pm$ 5.1%  | 101.3 $\pm$ 5.1 | 42.9 $\pm$ 3.7 | 33.10 $\pm$ 3.03 | 32.95 $\pm$ 3.25 | 33.1 $\pm$ 3.0 |
| DU outdoor  | 60   | 8   | V2  | 14.77 $\pm$ 2.18 | 25.30 $\pm$ 3.47 | 42.2% $\pm$ 5.8%  | 101.3 $\pm$ 5.1 | 38.2 $\pm$ 5.5 | 34.70 $\pm$ 3.47 | 23.79 $\pm$ 1.95 | 34.7 $\pm$ 3.5 |
| DU outdoor  | 60   | 10  | V1  | 16.20 $\pm$ 2.62 | 32.60 $\pm$ 4.99 | 54.3% $\pm$ 8.3%  | 101.3 $\pm$ 5.1 | 53.0 $\pm$ 6.6 | 27.40 $\pm$ 4.99 | 32.62 $\pm$ 2.58 | 27.4 $\pm$ 5.0 |
| DU outdoor  | 60   | 10  | V2  | 17.10 $\pm$ 3.07 | 30.70 $\pm$ 5.25 | 51.2% $\pm$ 8.8%  | 101.3 $\pm$ 5.1 | 47.8 $\pm$ 7.8 | 29.30 $\pm$ 5.25 | 25.73 $\pm$ 2.90 | 29.3 $\pm$ 5.3 |
| DU outdoor  | 90   | 3   | V1  | 6.26 $\pm$ 1.25  | 11.20 $\pm$ 1.69 | 12.4% $\pm$ 1.9%  | 151.2 $\pm$ 8.3 | 16.9 $\pm$ 2.3 | 78.80 $\pm$ 1.69 | 33.00 $\pm$ 3.00 | 78.8 $\pm$ 1.7 |
| DU outdoor  | 90   | 3   | V2  | 7.55 $\pm$ 1.47  | 10.90 $\pm$ 1.97 | 12.1% $\pm$ 2.2%  | 151.2 $\pm$ 8.3 | 14.1 $\pm$ 3.0 | 79.10 $\pm$ 1.97 | 20.23 $\pm$ 2.53 | 79.1 $\pm$ 2.0 |
| DU outdoor  | 90   | 5   | V1  | 9.80 $\pm$ 2.03  | 17.60 $\pm$ 2.88 | 19.6% $\pm$ 3.2%  | 151.2 $\pm$ 8.3 | 27.6 $\pm$ 4.3 | 72.40 $\pm$ 2.88 | 31.72 $\pm$ 2.92 | 72.4 $\pm$ 2.9 |
| DU outdoor  | 90   | 5   | V2  | 11.80 $\pm$ 2.02 | 17.50 $\pm$ 2.46 | 19.4% $\pm$ 2.7%  | 151.2 $\pm$ 8.3 | 23.9 $\pm$ 2.8 | 72.50 $\pm$ 2.46 | 19.99 $\pm$ 3.83 | 72.5 $\pm$ 2.5 |
| DU outdoor  | 90   | 8   | V1  | 14.08 $\pm$ 2.42 | 25.60 $\pm$ 3.50 | 28.4% $\pm$ 3.9%  | 151.2 $\pm$ 8.3 | 41.1 $\pm$ 5.5 | 64.40 $\pm$ 3.50 | 30.10 $\pm$ 2.84 | 64.4 $\pm$ 3.5 |
| DU outdoor  | 90   | 8   | V2  | 16.47 $\pm$ 2.44 | 25.50 $\pm$ 3.14 | 28.3% $\pm$ 3.5%  | 151.2 $\pm$ 8.3 | 37.1 $\pm$ 4.6 | 64.50 $\pm$ 3.14 | 20.81 $\pm$ 2.48 | 64.5 $\pm$ 3.1 |
| DU outdoor  | 90   | 10  | V1  | 16.76 $\pm$ 2.80 | 31.20 $\pm$ 4.24 | 34.7% $\pm$ 4.7%  | 151.2 $\pm$ 8.3 | 50.4 $\pm$ 6.7 | 58.80 $\pm$ 4.24 | 30.52 $\pm$ 3.05 | 58.8 $\pm$ 4.2 |
| DU outdoor  | 90   | 10  | V2  | 19.35 $\pm$ 2.56 | 31.10 $\pm$ 3.81 | 34.6% $\pm$ 4.2%  | 151.2 $\pm$ 8.3 | 47.0 $\pm$ 4.4 | 58.90 $\pm$ 3.81 | 21.94 $\pm$ 2.85 | 58.9 $\pm$ 3.8 |
| Synthetic   | 30   | 3   | V1  | 11.98 $\pm$ 1.48 | 22.60 $\pm$ 1.78 | 75.3% $\pm$ 5.9%  | 48.6 $\pm$ 3.6  | 34.5 $\pm$ 3.1 | 7.40 $\pm$ 1.78  | 23.66 $\pm$ 2.74 | 9.7 $\pm$ 1.3  |
| Synthetic   | 30   | 3   | V2  | 12.63 $\pm$ 1.29 | 22.00 $\pm$ 1.89 | 73.3% $\pm$ 6.3%  | 48.6 $\pm$ 3.6  | 32.7 $\pm$ 3.9 | 8.00 $\pm$ 1.89  | 20.22 $\pm$ 3.86 | 10.4 $\pm$ 1.2 |
| Synthetic   | 30   | 5   | V1  | 15.30 $\pm$ 1.10 | 27.20 $\pm$ 1.40 | 90.7% $\pm$ 4.7%  | 48.6 $\pm$ 3.6  | 43.8 $\pm$ 4.1 | 2.80 $\pm$ 1.40  | 16.15 $\pm$ 1.85 | 5.5 $\pm$ 1.4  |
| Synthetic   | 30   | 5   | V2  | 15.32 $\pm$ 0.99 | 27.00 $\pm$ 1.56 | 90.0% $\pm$ 5.2%  | 48.6 $\pm$ 3.6  | 43.2 $\pm$ 4.0 | 3.00 $\pm$ 1.56  | 16.66 $\pm$ 2.04 | 5.6 $\pm$ 1.3  |
| Synthetic   | 30   | 8   | V1  | 16.26 $\pm$ 1.14 | 27.50 $\pm$ 1.43 | 91.7% $\pm$ 4.8%  | 48.6 $\pm$ 3.6  | 44.2 $\pm$ 4.1 | 2.50 $\pm$ 1.43  | 11.41 $\pm$ 1.10 | 3.4 $\pm$ 1.1  |
| Synthetic   | 30   | 8   | V2  | 16.28 $\pm$ 1.11 | 27.50 $\pm$ 1.43 | 91.7% $\pm$ 4.8%  | 48.6 $\pm$ 3.6  | 44.2 $\pm$ 4.1 | 2.50 $\pm$ 1.43  | 11.38 $\pm$ 1.07 | 3.4 $\pm$ 1.1  |
| Synthetic   | 30   | 10  | V1  | 16.44 $\pm$ 1.12 | 27.50 $\pm$ 1.43 | 91.7% $\pm$ 4.8%  | 48.6 $\pm$ 3.6  | 44.2 $\pm$ 4.1 | 2.50 $\pm$ 1.43  | 10.37 $\pm$ 0.86 | 2.7 $\pm$ 1.3  |
| Synthetic   | 30   | 10  | V2  | 16.44 $\pm$ 1.12 | 27.50 $\pm$ 1.43 | 91.7% $\pm$ 4.8%  | 48.6 $\pm$ 3.6  | 44.2 $\pm$ 4.1 | 2.50 $\pm$ 1.43  | 10.37 $\pm$ 0.86 | 2.7 $\pm$ 1.3  |
| Synthetic   | 60   | 3   | V1  | 15.65 $\pm$ 1.86 | 25.40 $\pm$ 2.46 | 42.3% $\pm$ 4.1%  | 105.1 $\pm$ 6.7 | 39.1 $\pm$ 2.7 | 34.60 $\pm$ 2.46 | 21.60 $\pm$ 2.34 | 35.0 $\pm$ 2.2 |
| Synthetic   | 60   | 3   | V2  | 16.40 $\pm$ 1.62 | 24.70 $\pm$ 2.83 | 41.2% $\pm$ 4.7%  | 105.1 $\pm$ 6.7 | 36.2 $\pm$ 2.6 | 35.30 $\pm$ 2.83 | 16.31 $\pm$ 1.43 | 35.7 $\pm$ 2.6 |
| Synthetic   | 60   | 5   | V1  | 22.84 $\pm$ 2.17 | 39.90 $\pm$ 3.84 | 66.5% $\pm$ 6.4%  | 105.1 $\pm$ 6.7 | 63.9 $\pm$ 5.2 | 20.10 $\pm$ 3.84 | 21.64 $\pm$ 2.43 | 22.0 $\pm$ 2.7 |
| Synthetic   | 60   | 5   | V2  | 23.71 $\pm$ 1.69 | 38.60 $\pm$ 3.63 | 64.3% $\pm$ 6.0%  | 105.1 $\pm$ 6.7 | 60.7 $\pm$ 5.2 | 21.40 $\pm$ 3.63 | 18.03 $\pm$ 2.39 | 23.2 $\pm$ 2.8 |

| Environment | Task | UAV | Alg | Obj              | Served           | Task rate        | Req. parcel     | Drop parcel     | Deferred         | Delay            | Peak queue     |
|-------------|------|-----|-----|------------------|------------------|------------------|-----------------|-----------------|------------------|------------------|----------------|
| Synthetic   | 60   | 8   | V1  | 29.70 $\pm$ 2.66 | 53.30 $\pm$ 3.20 | 88.8% $\pm$ 5.3% | 105.1 $\pm$ 6.7 | 92.0 $\pm$ 7.2  | 6.70 $\pm$ 3.20  | 18.93 $\pm$ 3.11 | 11.0 $\pm$ 2.4 |
| Synthetic   | 60   | 8   | V2  | 29.78 $\pm$ 2.25 | 51.90 $\pm$ 3.90 | 86.5% $\pm$ 6.5% | 105.1 $\pm$ 6.7 | 88.7 $\pm$ 8.4  | 8.10 $\pm$ 3.90  | 18.63 $\pm$ 1.55 | 11.9 $\pm$ 2.6 |
| Synthetic   | 60   | 10  | V1  | 31.86 $\pm$ 1.72 | 55.30 $\pm$ 1.77 | 92.2% $\pm$ 2.9% | 105.1 $\pm$ 6.7 | 96.3 $\pm$ 7.7  | 4.70 $\pm$ 1.77  | 15.01 $\pm$ 2.54 | 8.7 $\pm$ 2.2  |
| Synthetic   | 60   | 10  | V2  | 31.77 $\pm$ 1.88 | 54.50 $\pm$ 2.59 | 90.8% $\pm$ 4.3% | 105.1 $\pm$ 6.7 | 94.2 $\pm$ 8.7  | 5.50 $\pm$ 2.59  | 15.17 $\pm$ 1.97 | 8.4 $\pm$ 2.4  |
| Synthetic   | 90   | 3   | V1  | 17.81 $\pm$ 1.59 | 27.70 $\pm$ 2.41 | 30.8% $\pm$ 2.7% | 156.4 $\pm$ 5.5 | 39.3 $\pm$ 3.1  | 62.30 $\pm$ 2.41 | 20.89 $\pm$ 2.28 | 62.3 $\pm$ 2.4 |
| Synthetic   | 90   | 3   | V2  | 18.34 $\pm$ 2.04 | 26.30 $\pm$ 3.09 | 29.2% $\pm$ 3.4% | 156.4 $\pm$ 5.5 | 35.7 $\pm$ 4.6  | 63.70 $\pm$ 3.09 | 14.87 $\pm$ 0.74 | 63.7 $\pm$ 3.1 |
| Synthetic   | 90   | 5   | V1  | 26.11 $\pm$ 2.29 | 41.80 $\pm$ 3.82 | 46.4% $\pm$ 4.2% | 156.4 $\pm$ 5.5 | 63.0 $\pm$ 5.2  | 48.20 $\pm$ 3.82 | 20.04 $\pm$ 2.16 | 48.5 $\pm$ 3.8 |
| Synthetic   | 90   | 5   | V2  | 26.73 $\pm$ 2.88 | 40.20 $\pm$ 4.44 | 44.7% $\pm$ 4.9% | 156.4 $\pm$ 5.5 | 59.4 $\pm$ 5.7  | 49.80 $\pm$ 4.44 | 15.95 $\pm$ 2.12 | 50.0 $\pm$ 4.3 |
| Synthetic   | 90   | 8   | V1  | 36.28 $\pm$ 2.87 | 62.30 $\pm$ 5.50 | 69.2% $\pm$ 6.1% | 156.4 $\pm$ 5.5 | 100.4 $\pm$ 9.3 | 27.70 $\pm$ 5.50 | 19.91 $\pm$ 1.98 | 29.3 $\pm$ 4.6 |
| Synthetic   | 90   | 8   | V2  | 37.00 $\pm$ 3.19 | 60.50 $\pm$ 5.38 | 67.2% $\pm$ 6.0% | 156.4 $\pm$ 5.5 | 95.6 $\pm$ 8.8  | 29.50 $\pm$ 5.38 | 17.08 $\pm$ 1.53 | 30.4 $\pm$ 5.0 |
| Synthetic   | 90   | 10  | V1  | 41.52 $\pm$ 2.75 | 74.30 $\pm$ 4.22 | 82.6% $\pm$ 4.7% | 156.4 $\pm$ 5.5 | 123.8 $\pm$ 8.0 | 15.70 $\pm$ 4.22 | 19.96 $\pm$ 2.31 | 19.6 $\pm$ 3.9 |
| Synthetic   | 90   | 10  | V2  | 41.65 $\pm$ 2.94 | 71.60 $\pm$ 5.10 | 79.6% $\pm$ 5.7% | 156.4 $\pm$ 5.5 | 118.1 $\pm$ 9.1 | 18.40 $\pm$ 5.10 | 18.33 $\pm$ 1.87 | 21.4 $\pm$ 4.6 |

সারণী 4: Resource metrics: mean  $\pm$  standard deviation।

| Environment | Task | UAV | Alg | Distance           | Travel           | Energy             | E/task            | E/parcel          | Safe return       |
|-------------|------|-----|-----|--------------------|------------------|--------------------|-------------------|-------------------|-------------------|
| DU outdoor  | 30   | 3   | V1  | 2832.4 $\pm$ 219.5 | 283.2 $\pm$ 21.9 | 2934.2 $\pm$ 224.0 | 299.0 $\pm$ 81.1  | 201.2 $\pm$ 37.0  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 30   | 3   | V2  | 2787.2 $\pm$ 184.9 | 278.7 $\pm$ 18.5 | 2887.1 $\pm$ 191.5 | 308.9 $\pm$ 113.1 | 254.2 $\pm$ 128.4 | 100.0% $\pm$ 0.0% |
| DU outdoor  | 30   | 5   | V1  | 4602.5 $\pm$ 327.9 | 460.3 $\pm$ 32.8 | 4762.7 $\pm$ 330.5 | 315.1 $\pm$ 87.1  | 213.3 $\pm$ 55.3  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 30   | 5   | V2  | 4542.6 $\pm$ 271.2 | 454.3 $\pm$ 27.1 | 4722.1 $\pm$ 290.6 | 319.9 $\pm$ 77.4  | 239.7 $\pm$ 82.0  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 30   | 8   | V1  | 6495.0 $\pm$ 350.3 | 649.5 $\pm$ 35.0 | 6709.3 $\pm$ 357.3 | 342.0 $\pm$ 73.6  | 225.6 $\pm$ 55.4  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 30   | 8   | V2  | 6516.3 $\pm$ 384.0 | 651.6 $\pm$ 38.4 | 6748.6 $\pm$ 404.1 | 358.3 $\pm$ 85.7  | 245.7 $\pm$ 90.0  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 30   | 10  | V1  | 7404.9 $\pm$ 529.3 | 740.5 $\pm$ 52.9 | 7635.4 $\pm$ 542.9 | 367.6 $\pm$ 70.1  | 241.2 $\pm$ 53.2  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 30   | 10  | V2  | 7380.9 $\pm$ 625.2 | 738.1 $\pm$ 62.5 | 7624.9 $\pm$ 647.6 | 373.6 $\pm$ 65.9  | 251.0 $\pm$ 64.7  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 60   | 3   | V1  | 2942.4 $\pm$ 162.8 | 294.2 $\pm$ 16.3 | 3058.2 $\pm$ 165.8 | 290.4 $\pm$ 54.4  | 184.9 $\pm$ 23.1  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 60   | 3   | V2  | 2929.5 $\pm$ 250.8 | 292.9 $\pm$ 25.1 | 3057.3 $\pm$ 252.9 | 276.1 $\pm$ 62.3  | 216.6 $\pm$ 47.4  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 60   | 5   | V1  | 4751.1 $\pm$ 265.1 | 475.1 $\pm$ 26.5 | 4939.5 $\pm$ 270.9 | 281.6 $\pm$ 43.5  | 178.3 $\pm$ 20.8  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 60   | 5   | V2  | 4782.8 $\pm$ 273.1 | 478.3 $\pm$ 27.3 | 4981.2 $\pm$ 271.4 | 297.6 $\pm$ 63.5  | 215.0 $\pm$ 34.8  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 60   | 8   | V1  | 7604.1 $\pm$ 342.3 | 760.4 $\pm$ 34.2 | 7897.7 $\pm$ 357.3 | 297.5 $\pm$ 40.9  | 185.3 $\pm$ 17.5  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 60   | 8   | V2  | 7373.2 $\pm$ 323.8 | 737.3 $\pm$ 32.4 | 7667.6 $\pm$ 331.0 | 309.0 $\pm$ 49.5  | 204.8 $\pm$ 32.6  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 60   | 10  | V1  | 9321.8 $\pm$ 522.4 | 932.2 $\pm$ 52.2 | 9681.0 $\pm$ 538.8 | 303.4 $\pm$ 49.8  | 184.9 $\pm$ 21.9  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 60   | 10  | V2  | 9162.9 $\pm$ 461.7 | 916.3 $\pm$ 46.2 | 9526.9 $\pm$ 492.9 | 318.7 $\pm$ 55.3  | 204.0 $\pm$ 32.9  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 90   | 3   | V1  | 3038.9 $\pm$ 122.2 | 303.9 $\pm$ 12.2 | 3153.7 $\pm$ 123.5 | 287.2 $\pm$ 43.1  | 189.7 $\pm$ 25.4  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 90   | 3   | V2  | 2883.3 $\pm$ 189.1 | 288.3 $\pm$ 18.9 | 2998.0 $\pm$ 192.0 | 284.4 $\pm$ 61.1  | 221.7 $\pm$ 50.5  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 90   | 5   | V1  | 4960.8 $\pm$ 186.1 | 496.1 $\pm$ 18.6 | 5141.6 $\pm$ 183.6 | 300.1 $\pm$ 55.8  | 190.9 $\pm$ 33.9  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 90   | 5   | V2  | 4918.5 $\pm$ 233.8 | 491.8 $\pm$ 23.4 | 5110.3 $\pm$ 218.9 | 298.0 $\pm$ 49.2  | 216.9 $\pm$ 30.3  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 90   | 8   | V1  | 7667.8 $\pm$ 377.4 | 766.8 $\pm$ 37.7 | 7941.7 $\pm$ 381.3 | 316.3 $\pm$ 51.7  | 196.8 $\pm$ 30.8  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 90   | 8   | V2  | 7709.9 $\pm$ 290.4 | 771.0 $\pm$ 29.0 | 8005.1 $\pm$ 298.4 | 318.2 $\pm$ 39.7  | 218.4 $\pm$ 24.8  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 90   | 10  | V1  | 9516.9 $\pm$ 390.9 | 951.7 $\pm$ 39.1 | 9854.2 $\pm$ 393.7 | 321.9 $\pm$ 50.1  | 198.8 $\pm$ 28.2  | 100.0% $\pm$ 0.0% |
| DU outdoor  | 90   | 10  | V2  | 9262.3 $\pm$ 497.4 | 926.2 $\pm$ 49.7 | 9614.5 $\pm$ 492.4 | 313.8 $\pm$ 44.5  | 206.2 $\pm$ 21.1  | 100.0% $\pm$ 0.0% |
| Synthetic   | 30   | 3   | V1  | 2521.9 $\pm$ 135.6 | 252.2 $\pm$ 13.6 | 2671.2 $\pm$ 135.9 | 118.7 $\pm$ 8.8   | 78.0 $\pm$ 7.9    | 100.0% $\pm$ 0.0% |
| Synthetic   | 30   | 3   | V2  | 2607.0 $\pm$ 131.7 | 260.7 $\pm$ 13.2 | 2765.2 $\pm$ 132.2 | 126.4 $\pm$ 10.7  | 85.7 $\pm$ 10.9   | 100.0% $\pm$ 0.0% |
| Synthetic   | 30   | 5   | V1  | 3801.4 $\pm$ 301.6 | 380.1 $\pm$ 30.2 | 4001.2 $\pm$ 310.4 | 147.3 $\pm$ 12.1  | 92.2 $\pm$ 12.0   | 100.0% $\pm$ 0.0% |
| Synthetic   | 30   | 5   | V2  | 3863.4 $\pm$ 277.9 | 386.3 $\pm$ 27.8 | 4071.7 $\pm$ 286.8 | 151.0 $\pm$ 10.4  | 94.9 $\pm$ 10.5   | 100.0% $\pm$ 0.0% |
| Synthetic   | 30   | 8   | V1  | 4766.2 $\pm$ 471.5 | 476.6 $\pm$ 47.1 | 4980.5 $\pm$ 484.6 | 181.3 $\pm$ 17.5  | 113.7 $\pm$ 16.2  | 100.0% $\pm$ 0.0% |
| Synthetic   | 30   | 8   | V2  | 4786.7 $\pm$ 479.4 | 478.7 $\pm$ 47.9 | 5002.2 $\pm$ 492.8 | 182.1 $\pm$ 17.7  | 114.2 $\pm$ 16.3  | 100.0% $\pm$ 0.0% |
| Synthetic   | 30   | 10  | V1  | 4992.2 $\pm$ 538.4 | 499.2 $\pm$ 53.8 | 5209.9 $\pm$ 552.0 | 189.7 $\pm$ 20.1  | 119.0 $\pm$ 17.8  | 100.0% $\pm$ 0.0% |
| Synthetic   | 30   | 10  | V2  | 4992.2 $\pm$ 538.4 | 499.2 $\pm$ 53.8 | 5209.9 $\pm$ 552.0 | 189.7 $\pm$ 20.1  | 119.0 $\pm$ 17.8  | 100.0% $\pm$ 0.0% |
| Synthetic   | 60   | 3   | V1  | 2685.7 $\pm$ 81.4  | 268.6 $\pm$ 8.1  | 2842.1 $\pm$ 83.0  | 112.8 $\pm$ 10.5  | 73.0 $\pm$ 6.2    | 100.0% $\pm$ 0.0% |
| Synthetic   | 60   | 3   | V2  | 2706.9 $\pm$ 72.6  | 270.7 $\pm$ 7.3  | 2866.8 $\pm$ 78.6  | 117.9 $\pm$ 17.5  | 79.6 $\pm$ 7.1    | 100.0% $\pm$ 0.0% |
| Synthetic   | 60   | 5   | V1  | 4406.2 $\pm$ 135.9 | 440.6 $\pm$ 13.6 | 4666.6 $\pm$ 140.4 | 118.0 $\pm$ 12.2  | 73.4 $\pm$ 5.7    | 100.0% $\pm$ 0.0% |
| Synthetic   | 60   | 5   | V2  | 4444.6 $\pm$ 128.9 | 444.5 $\pm$ 12.9 | 4708.6 $\pm$ 140.9 | 123.0 $\pm$ 12.6  | 78.0 $\pm$ 5.3    | 100.0% $\pm$ 0.0% |
| Synthetic   | 60   | 8   | V1  | 6587.8 $\pm$ 374.3 | 658.8 $\pm$ 37.4 | 6966.9 $\pm$ 394.9 | 131.2 $\pm$ 12.0  | 76.2 $\pm$ 7.5    | 100.0% $\pm$ 0.0% |

| Environment | Task | UAV | Alg | Distance           | Travel           | Energy             | E/task           | E/parcel        | Safe return       |
|-------------|------|-----|-----|--------------------|------------------|--------------------|------------------|-----------------|-------------------|
| Synthetic   | 60   | 8   | V2  | 6670.9 $\pm$ 238.7 | 667.1 $\pm$ 23.9 | 7065.6 $\pm$ 255.6 | 137.1 $\pm$ 15.0 | 80.5 $\pm$ 9.8  | 100.0% $\pm$ 0.0% |
| Synthetic   | 60   | 10  | V1  | 7628.7 $\pm$ 491.4 | 762.9 $\pm$ 49.1 | 8037.7 $\pm$ 523.9 | 145.5 $\pm$ 11.0 | 83.8 $\pm$ 7.3  | 100.0% $\pm$ 0.0% |
| Synthetic   | 60   | 10  | V2  | 7647.2 $\pm$ 525.5 | 764.7 $\pm$ 52.5 | 8073.4 $\pm$ 561.6 | 148.5 $\pm$ 13.7 | 86.3 $\pm$ 9.1  | 100.0% $\pm$ 0.0% |
| Synthetic   | 90   | 3   | V1  | 2778.2 $\pm$ 70.1  | 277.8 $\pm$ 7.0  | 2943.9 $\pm$ 67.6  | 107.1 $\pm$ 10.5 | 75.4 $\pm$ 7.0  | 100.0% $\pm$ 0.0% |
| Synthetic   | 90   | 3   | V2  | 2737.4 $\pm$ 64.2  | 273.7 $\pm$ 6.4  | 2898.1 $\pm$ 66.4  | 111.7 $\pm$ 14.9 | 82.7 $\pm$ 13.5 | 100.0% $\pm$ 0.0% |
| Synthetic   | 90   | 5   | V1  | 4541.3 $\pm$ 129.3 | 454.1 $\pm$ 12.9 | 4804.2 $\pm$ 130.8 | 115.9 $\pm$ 12.1 | 76.8 $\pm$ 8.0  | 100.0% $\pm$ 0.0% |
| Synthetic   | 90   | 5   | V2  | 4552.5 $\pm$ 130.1 | 455.2 $\pm$ 13.0 | 4824.7 $\pm$ 144.8 | 121.6 $\pm$ 15.9 | 82.1 $\pm$ 10.0 | 100.0% $\pm$ 0.0% |
| Synthetic   | 90   | 8   | V1  | 7056.7 $\pm$ 176.0 | 705.7 $\pm$ 17.6 | 7472.0 $\pm$ 182.3 | 120.7 $\pm$ 9.5  | 75.0 $\pm$ 6.6  | 100.0% $\pm$ 0.0% |
| Synthetic   | 90   | 8   | V2  | 7171.5 $\pm$ 53.5  | 717.1 $\pm$ 5.4  | 7598.7 $\pm$ 57.4  | 126.6 $\pm$ 12.0 | 80.1 $\pm$ 8.1  | 100.0% $\pm$ 0.0% |
| Synthetic   | 90   | 10  | V1  | 8730.8 $\pm$ 238.3 | 873.1 $\pm$ 23.8 | 9240.4 $\pm$ 251.5 | 124.8 $\pm$ 8.5  | 75.0 $\pm$ 5.9  | 100.0% $\pm$ 0.0% |
| Synthetic   | 90   | 10  | V2  | 8823.4 $\pm$ 197.4 | 882.3 $\pm$ 19.7 | 9340.6 $\pm$ 217.6 | 131.0 $\pm$ 9.7  | 79.5 $\pm$ 6.2  | 100.0% $\pm$ 0.0% |

সারণী 5: Recourse এবং runtime metrics: mean  $\pm$  standard deviation।

| Environment | Task | UAV | Alg | Trigger        | Accepted       | Rejected       | Gain              | Runtime              |
|-------------|------|-----|-----|----------------|----------------|----------------|-------------------|----------------------|
| DU outdoor  | 30   | 3   | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 11.485 $\pm$ 6.489   |
| DU outdoor  | 30   | 3   | V2  | 20.2 $\pm$ 2.7 | 6.1 $\pm$ 2.4  | 14.1 $\pm$ 1.5 | 1.302 $\pm$ 0.573 | 37.612 $\pm$ 26.444  |
| DU outdoor  | 30   | 5   | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 13.509 $\pm$ 7.659   |
| DU outdoor  | 30   | 5   | V2  | 15.2 $\pm$ 2.6 | 7.0 $\pm$ 1.9  | 8.2 $\pm$ 2.8  | 1.409 $\pm$ 0.380 | 42.588 $\pm$ 22.340  |
| DU outdoor  | 30   | 8   | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 15.143 $\pm$ 8.093   |
| DU outdoor  | 30   | 8   | V2  | 6.9 $\pm$ 3.1  | 4.7 $\pm$ 1.7  | 2.2 $\pm$ 2.1  | 0.903 $\pm$ 0.461 | 38.111 $\pm$ 19.065  |
| DU outdoor  | 30   | 10  | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 14.422 $\pm$ 7.936   |
| DU outdoor  | 30   | 10  | V2  | 5.1 $\pm$ 3.1  | 3.2 $\pm$ 1.3  | 1.9 $\pm$ 2.3  | 0.676 $\pm$ 0.347 | 28.559 $\pm$ 12.537  |
| DU outdoor  | 60   | 3   | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 12.684 $\pm$ 6.289   |
| DU outdoor  | 60   | 3   | V2  | 43.8 $\pm$ 4.4 | 8.0 $\pm$ 1.9  | 35.8 $\pm$ 4.8 | 1.350 $\pm$ 0.429 | 50.711 $\pm$ 34.337  |
| DU outdoor  | 60   | 5   | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 18.059 $\pm$ 7.983   |
| DU outdoor  | 60   | 5   | V2  | 36.0 $\pm$ 4.9 | 9.1 $\pm$ 1.9  | 26.9 $\pm$ 5.2 | 1.660 $\pm$ 0.435 | 54.161 $\pm$ 25.153  |
| DU outdoor  | 60   | 8   | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 23.971 $\pm$ 10.611  |
| DU outdoor  | 60   | 8   | V2  | 30.3 $\pm$ 5.5 | 13.0 $\pm$ 2.3 | 17.3 $\pm$ 4.7 | 2.283 $\pm$ 0.365 | 50.260 $\pm$ 25.969  |
| DU outdoor  | 60   | 10  | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 30.386 $\pm$ 14.888  |
| DU outdoor  | 60   | 10  | V2  | 26.8 $\pm$ 4.8 | 13.0 $\pm$ 2.1 | 13.8 $\pm$ 3.9 | 2.599 $\pm$ 0.624 | 36.383 $\pm$ 17.980  |
| DU outdoor  | 90   | 3   | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 15.823 $\pm$ 9.331   |
| DU outdoor  | 90   | 3   | V2  | 66.9 $\pm$ 5.1 | 9.7 $\pm$ 3.1  | 57.2 $\pm$ 5.9 | 1.369 $\pm$ 0.524 | 65.505 $\pm$ 38.326  |
| DU outdoor  | 90   | 5   | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 20.151 $\pm$ 8.642   |
| DU outdoor  | 90   | 5   | V2  | 59.6 $\pm$ 4.9 | 14.0 $\pm$ 5.1 | 45.6 $\pm$ 8.6 | 2.157 $\pm$ 0.889 | 61.833 $\pm$ 25.722  |
| DU outdoor  | 90   | 8   | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 30.704 $\pm$ 12.132  |
| DU outdoor  | 90   | 8   | V2  | 48.8 $\pm$ 3.9 | 17.4 $\pm$ 4.1 | 31.4 $\pm$ 6.0 | 2.923 $\pm$ 0.867 | 113.128 $\pm$ 49.163 |
| DU outdoor  | 90   | 10  | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 36.365 $\pm$ 8.356   |
| DU outdoor  | 90   | 10  | V2  | 40.5 $\pm$ 6.3 | 17.1 $\pm$ 2.7 | 23.4 $\pm$ 6.4 | 3.208 $\pm$ 0.810 | 134.039 $\pm$ 47.750 |
| Synthetic   | 30   | 3   | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 2.069 $\pm$ 1.400    |
| Synthetic   | 30   | 3   | V2  | 16.9 $\pm$ 2.2 | 6.4 $\pm$ 2.0  | 10.5 $\pm$ 3.0 | 1.627 $\pm$ 0.672 | 4.040 $\pm$ 2.335    |
| Synthetic   | 30   | 5   | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 1.118 $\pm$ 0.569    |
| Synthetic   | 30   | 5   | V2  | 8.0 $\pm$ 3.1  | 3.7 $\pm$ 1.3  | 4.3 $\pm$ 2.7  | 0.876 $\pm$ 0.444 | 3.222 $\pm$ 1.809    |
| Synthetic   | 30   | 8   | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 1.102 $\pm$ 0.597    |
| Synthetic   | 30   | 8   | V2  | 1.4 $\pm$ 1.0  | 0.8 $\pm$ 1.0  | 0.6 $\pm$ 0.7  | 0.159 $\pm$ 0.281 | 3.355 $\pm$ 1.723    |
| Synthetic   | 30   | 10  | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 1.078 $\pm$ 0.540    |
| Synthetic   | 30   | 10  | V2  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 3.612 $\pm$ 1.897    |
| Synthetic   | 60   | 3   | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 2.862 $\pm$ 1.195    |
| Synthetic   | 60   | 3   | V2  | 34.3 $\pm$ 4.6 | 8.8 $\pm$ 3.1  | 25.5 $\pm$ 3.6 | 1.532 $\pm$ 0.637 | 11.164 $\pm$ 4.674   |
| Synthetic   | 60   | 5   | V1  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0  | 0.000 $\pm$ 0.000 | 4.039 $\pm$ 1.683    |
| Synthetic   | 60   | 5   | V2  | 25.1 $\pm$ 5.7 | 11.0 $\pm$ 2.9 | 14.1 $\pm$ 3.9 | 2.320 $\pm$ 0.935 | 14.512 $\pm$ 4.221   |

| Environment | Task | UAV | Alg | Trigger         | Accepted       | Rejected       | Gain              | Runtime            |
|-------------|------|-----|-----|-----------------|----------------|----------------|-------------------|--------------------|
| Synthetic   | 60   | 8   | V1  | $0.0 \pm 0.0$   | $0.0 \pm 0.0$  | $0.0 \pm 0.0$  | $0.000 \pm 0.000$ | $4.663 \pm 3.470$  |
| Synthetic   | 60   | 8   | V2  | $14.8 \pm 5.7$  | $9.7 \pm 3.3$  | $5.1 \pm 3.3$  | $2.466 \pm 1.039$ | $17.878 \pm 7.983$ |
| Synthetic   | 60   | 10  | V1  | $0.0 \pm 0.0$   | $0.0 \pm 0.0$  | $0.0 \pm 0.0$  | $0.000 \pm 0.000$ | $3.644 \pm 2.618$  |
| Synthetic   | 60   | 10  | V2  | $8.5 \pm 4.6$   | $5.4 \pm 3.0$  | $3.1 \pm 3.0$  | $1.472 \pm 0.938$ | $11.775 \pm 5.478$ |
| Synthetic   | 90   | 3   | V1  | $0.0 \pm 0.0$   | $0.0 \pm 0.0$  | $0.0 \pm 0.0$  | $0.000 \pm 0.000$ | $3.246 \pm 1.557$  |
| Synthetic   | 90   | 3   | V2  | $57.9 \pm 4.6$  | $13.3 \pm 2.4$ | $44.6 \pm 4.9$ | $1.906 \pm 0.364$ | $8.992 \pm 4.750$  |
| Synthetic   | 90   | 5   | V1  | $0.0 \pm 0.0$   | $0.0 \pm 0.0$  | $0.0 \pm 0.0$  | $0.000 \pm 0.000$ | $5.201 \pm 3.266$  |
| Synthetic   | 90   | 5   | V2  | $44.1 \pm 7.1$  | $15.5 \pm 4.2$ | $28.6 \pm 5.3$ | $2.442 \pm 0.854$ | $11.353 \pm 5.917$ |
| Synthetic   | 90   | 8   | V1  | $0.0 \pm 0.0$   | $0.0 \pm 0.0$  | $0.0 \pm 0.0$  | $0.000 \pm 0.000$ | $7.286 \pm 2.764$  |
| Synthetic   | 90   | 8   | V2  | $28.7 \pm 6.4$  | $14.4 \pm 4.1$ | $14.3 \pm 4.1$ | $2.787 \pm 1.140$ | $16.359 \pm 9.764$ |
| Synthetic   | 90   | 10  | V1  | $0.0 \pm 0.0$   | $0.0 \pm 0.0$  | $0.0 \pm 0.0$  | $0.000 \pm 0.000$ | $8.488 \pm 3.596$  |
| Synthetic   | 90   | 10  | V2  | $24.1 \pm 10.2$ | $13.8 \pm 6.5$ | $10.3 \pm 4.7$ | $3.236 \pm 1.573$ | $17.993 \pm 7.594$ |

এই expanded matrix-এ absolute requested/dropped parcel, task rate, deferred task এবং peak queue সরাসরি দেওয়া হয়েছে; parcel rate একা দেখে service volume অনুমান করতে হবে না। Resource table-এ distance এবং energy পাশাপাশি থাকায় route length বাড়ার সঙ্গে payload, ascent এবং service energy কীভাবে বদলেছে সেটিও দেখা যায়। Recourse table-এ trigger, accepted/rejected replacement, gain এবং runtime আলাদা করে দেখানো হয়েছে।

## 12 Recourse evidence: environment, load, fleet এবং phase

Trigger-level মোট outcome-এর পাশাপাশি trace থেকে acceptance rate আলাদা করে হিসাব করা হয়েছে। Accepted gain-এর mean  $\pm$  standard deviation এখানে accepted event-এর  $\Delta J$ ; episode-level *replacement\_gain* তার থেকে আলাদা cumulative quantity।

| Group type    | Group      | Trigger | Accepted | Rate  | Rejected | Gain mean $\pm$ std |
|---------------|------------|---------|----------|-------|----------|---------------------|
| Environment   | DU outdoor | 4001    | 1223     | 30.6% | 2778     | 0.179 $\pm$ 0.132   |
| Environment   | Synthetic  | 2638    | 1028     | 39.0% | 1610     | 0.203 $\pm$ 0.150   |
| Task load     | 30         | 737     | 319      | 43.3% | 418      | 0.218 $\pm$ 0.141   |
| Task load     | 60         | 2196    | 780      | 35.5% | 1416     | 0.201 $\pm$ 0.146   |
| Task load     | 90         | 3706    | 1152     | 31.1% | 2554     | 0.174 $\pm$ 0.136   |
| UAV count     | 3          | 2400    | 523      | 21.8% | 1877     | 0.174 $\pm$ 0.134   |
| UAV count     | 5          | 1880    | 603      | 32.1% | 1277     | 0.180 $\pm$ 0.133   |
| UAV count     | 8          | 1309    | 600      | 45.8% | 709      | 0.192 $\pm$ 0.139   |
| UAV count     | 10         | 1050    | 525      | 50.0% | 525      | 0.213 $\pm$ 0.155   |
| Arrival phase | High       | 3728    | 1223     | 32.8% | 2505     | 0.173 $\pm$ 0.130   |
| Arrival phase | Medium     | 2265    | 886      | 39.1% | 1379     | 0.201 $\pm$ 0.143   |
| Arrival phase | Low        | 646     | 142      | 22.0% | 504      | 0.255 $\pm$ 0.191   |

12.1 Candidate-level rejection breakdown একটি trigger-এর ভিতরে একাধিক candidate থাকতে পারে। তাই নিচের count candidate-level; আগের rejection table-এর count trigger-level final reason।

| Candidate outcome/reason | Count | Share |
|--------------------------|-------|-------|
| Onboard inventory        | 21828 | 62.6% |
| Payload capacity         | 0     | 0.0%  |
| Route infeasible         | 5     | 0.0%  |
| Energy reserve           | 154   | 0.4%  |
| Horizon                  | 851   | 2.4%  |
| Service in progress      | 1447  | 4.1%  |
| Non-positive gain        | 6021  | 17.3% |
| Feasible positive gain   | 4584  | 13.1% |

Accepted replacement-এর পরে 2025টি unique displaced-task instance-এর মধ্যে 558টি (27.6%) horizon-এর মধ্যে পরে complete হয়েছে এবং 1467টি (72.4%) deferred থেকেছে। মোট accepted event ছিল 2251; এর মধ্যে 226টি repeated displacement event, তাই event count এবং unique task count আলাদা করে report করা হয়েছে। Accepted trace example: du\_outdoor\_tasks\_30\_uavs\_3\_seed\_101-এ  $t = 7.78$  min-এ uav\_1 দ্বারা task\_6 task-কে সরিয়ে task\_10 task বসানো হয়েছে; old value 0.244, new value 0.633, gain +0.390. Rejected trace example: du\_outdoor\_tasks\_30\_uavs\_3\_seed\_101-এ  $t = 15.10$  min-এ task\_9 task-এর জন্য candidate gain -0.076; candidate feasible হলেও gain positive নয়, তাই final reason no\_positive\_gain এবং task queue-তে রাখা হয়েছে।

## 13 Validation evidence

একটি sentence-এর বদলে paired matrix এবং ছোট deterministic replay fixture-এর evidence নিচে দেওয়া হলো। Fixed-control fixture-এ V1 এবং V2 recourse-disabled একই scenario-তে চালানো হয়েছে; heavy matrix-এর ২৪০টি pair-এ scenario fingerprint match আলাদা করে যাচাই করা হয়েছে।

| Validation check                               | Evidence                                                | Result |
|------------------------------------------------|---------------------------------------------------------|--------|
| Same scenario fingerprint এবং task realization | 240টি paired cell-এর fingerprint match                  | PASS   |
| Fixed-control objective                        | V1 5.453757; V2 fixed 5.453757                          | PASS   |
| Fixed-control served task                      | Same served-task ID set                                 | PASS   |
| Fixed-control distance and energy              | Both values match within tolerance                      | PASS   |
| Fixed-control candidate count                  | Same candidate count                                    | PASS   |
| Deterministic replay                           | Repeated V2 run produced identical events and decisions | PASS   |
| Duplicate completion                           | Completed event IDs are unique                          | PASS   |
| Objective double-counting                      | Reported 5.453757; direct final-completion sum 5.453757 | PASS   |

Validation fixture: synthetic, seed 5, 12 tasks, 3 UAV. এটি report-এর reproducibility check; full result matrix-এর performance estimate নয়।

## 14 V2 result-এর অর্থ এবং limitation

1. V2 active mission-এর uncompleted suffix পরিবর্তন করে নতুন task-এর priority value বিবেচনা করেছে।
2. Accepted replacement-এ displaced task queue-তে ফিরে যায়; task হারিয়ে যায় না।
3. Current onboard inventory না মিললে V2 replacement গ্রহণ করে না।
4. Completed route prefix এবং completed delivery immutable রাখা হয়েছে।
5. Final objective completed task-এর service value থেকে একবার গণনা করা হয়েছে।
6. Safe return rate এবং energy value configured simulation model-এর ফল, field flight reliability নয়।
7. One-for-one V2 full mathematical optimizer বা full fleet rolling-horizon optimizer নয়।
8. Reserve inventory, base reload, multiple replacement এবং broader reassignment V3 scope।

## 15 উপসংহার এবং পরবর্তী কাজ

V1 একটি fixed active-mission reference হিসেবে অপরিবর্তিত রাখা হয়েছে। V2 formulation-এর সীমিত recourse rule যোগ করেছে: নতুন high-value task এলে feasible হলে একটি active mission-এর একটি uncompleted task replace করা যায়। Paired matrix V1 এবং V2-এর operational outcome এবং recourse overhead একসঙ্গে দেখাবে। পরবর্তী version-এ reserve inventory এবং active inventory-aware reassignment যোগ করার আগে V2-এর decision trace, service-value gain এবং runtime overhead বিশ্লেষণ করা হবে।

## 16 Reproducibility artifacts

- Simulation/results/pgbm\_v2\_recourse\_experiments/raw\_metrics/pgbm\_v2\_paired\_matrix\_metrics\_tasks\_30\_60\_90\_uavs\_3\_5\_8\_10\_seeds\_101\_110.csv: 8৮০টি metrics row।
- Simulation/results/pgbm\_v2\_recourse\_experiments/decision\_traces/pgbm\_v2\_decision\_traces.jsonl: ২৪০টি V2 decision trace।
- Simulation/run\_pgbm\_v2\_experiment\_matrix.py: matrix execution script।
- Simulation/pgbm\_sim/v2/: V2 runtime, recourse এবং experiment modules।
- docs/specs/0007-pgbm-v2-task-replacement-recourse.md: accepted design specification।